ENERGYCHAIN

THE PROBLEMS WE AIM TO SOLVE

India's energy and carbon markets are growing fast — but the rules underneath them are still broken.

Too Many Middlemen

Multiple intermediaries sit between buyer and seller, making every electricity and carbon-credit trade slower and more expensive.

Prosumers Locked Out

India has 30+ GW of rooftop solar installed, but small producers can only sell back to the DISCOM at a fixed tariff — never directly to a neighbour.

Opaque Carbon Credits

The market runs on trust in registries, not verifiable proof — leaving room for double-counting, unretired credits, and greenwashing.

Safety & Trust Gaps

There's no way to verify a counterparty or a credit's legitimacy without exposing sensitive business or usage data in the process.

WHY IT MATTERS

30 GW+

Rooftop solar installed in India (May 2026)

1 Crore

Household solar target under PM Surya Ghar

$2.29B → $4.92B

Voluntary carbon credit market growth, by 2030

OUR SOLUTION

VERIFICATION AT THE SOURCE

- Phone camera scans the physical meter, OCR reads it on-device
- No self-declared numbers — the ledger is only as honest as its input
- Solves the "garbage-in" flaw every competing platform ignores

TWO ASSETS, ONE CLICK

- Smart contract mints carbon credits the moment energy is listed
- Electricity + carbon credit settle in a single transaction
- Prosumers get paid twice for the same clean unit

PRIVATE, YET PROVABLE

- ZK-SNARK circuits (Circom + snarkjs) prove "I have enough" without revealing balance, price or usage
- Public verifiability with zero data leakage
- Makes the platform usable by businesses, not just hobbyists

ANCHORED, NOT DUMPED

- Meter data → IPFS; only the hash goes on-chain
- Tamper-evident records at negligible gas cost
- Scales from one rooftop to an entire DISCOM
- Prove it later - anyone can re-fetch the original reading from the hash and confirm it was never edited

TECH STACKS

4
Core Technical Layers

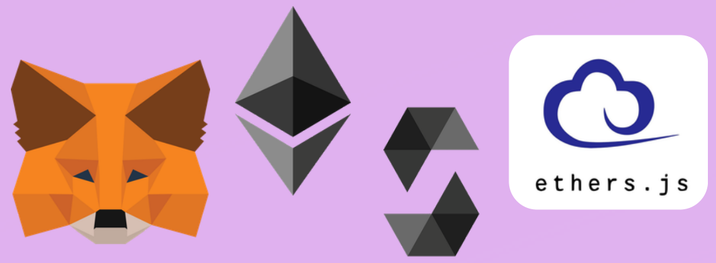
100%
Decentralisation Storage Stack

ZERO
Knowledge Proof Integration (zk SNARK)

FRONTEND UI STACK



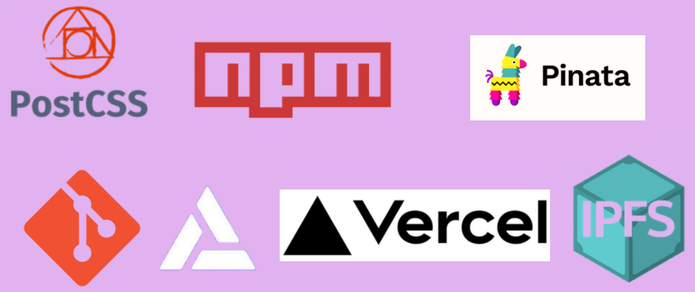
BLOCKCHAIN & WEB3 LAYER



BACKEND & PRIVACY



INFRA & STORAGE





System architecture

Frontend interaction

- Metamask connnect window
- Trading window
  - Energy Dashboard
    - Current Generation
    - Consumption
    - Carbon Credits
  - Trading windows
    - Carbon Credit Trading
    - Energy Trading

Based on emissions reductions, CarbonCred and ElectroCred tokens are minted and assigned to factory owners.

The backend acts as the processing hub, connecting the frontend, blockchain, IoT data sources, and ZK-SNARKs.

Backend logic

Displays data such as pollution index, token prices, number of users, etc., for different regions

- Real-Time Data Processing
- User data management
- Smart contracts
  - Energy Trading
  - Carbon Trading
  - IOT Integration

IoT and Emission Data Integration

- Data validation
- Data flow
- Token generation

Blockchain integration

- Ledger storage
- Smart contracts
  - Token minting & burning
  - Automated trading and settlement
  - Prevent double-spending

Deployed on a scalable blockchain (e.g., Ethereum Layer 2 or Polygon)

ZK-SNARKs for Privacy

The blockchain smart contract includes verification logic for ZK-SNARK proofs.

- Integration points
- Use cases
  - Trade privacy
  - Emission data privacy

Proves that a user has sufficient tokens for a trade without revealing their exact balance

IOT INTEGRATION FLOW

DATA COLLECTION & OCR READER

- IoT devices and smart meters continuously monitor and collect data.
- This data is formatted and saved into a JSON file.

STORING DATA ON IPFS

- The JSON file is uploaded to the InterPlanetary File System (IPFS).
- A unique Content Identifier (CID) or hash is generated after successful storage.

STORING THE HASH ON BLOCKCHAIN

- The generated CID is stored on the blockchain.
- We utilize Alchemy to handle blockchain transactions and storage operations.

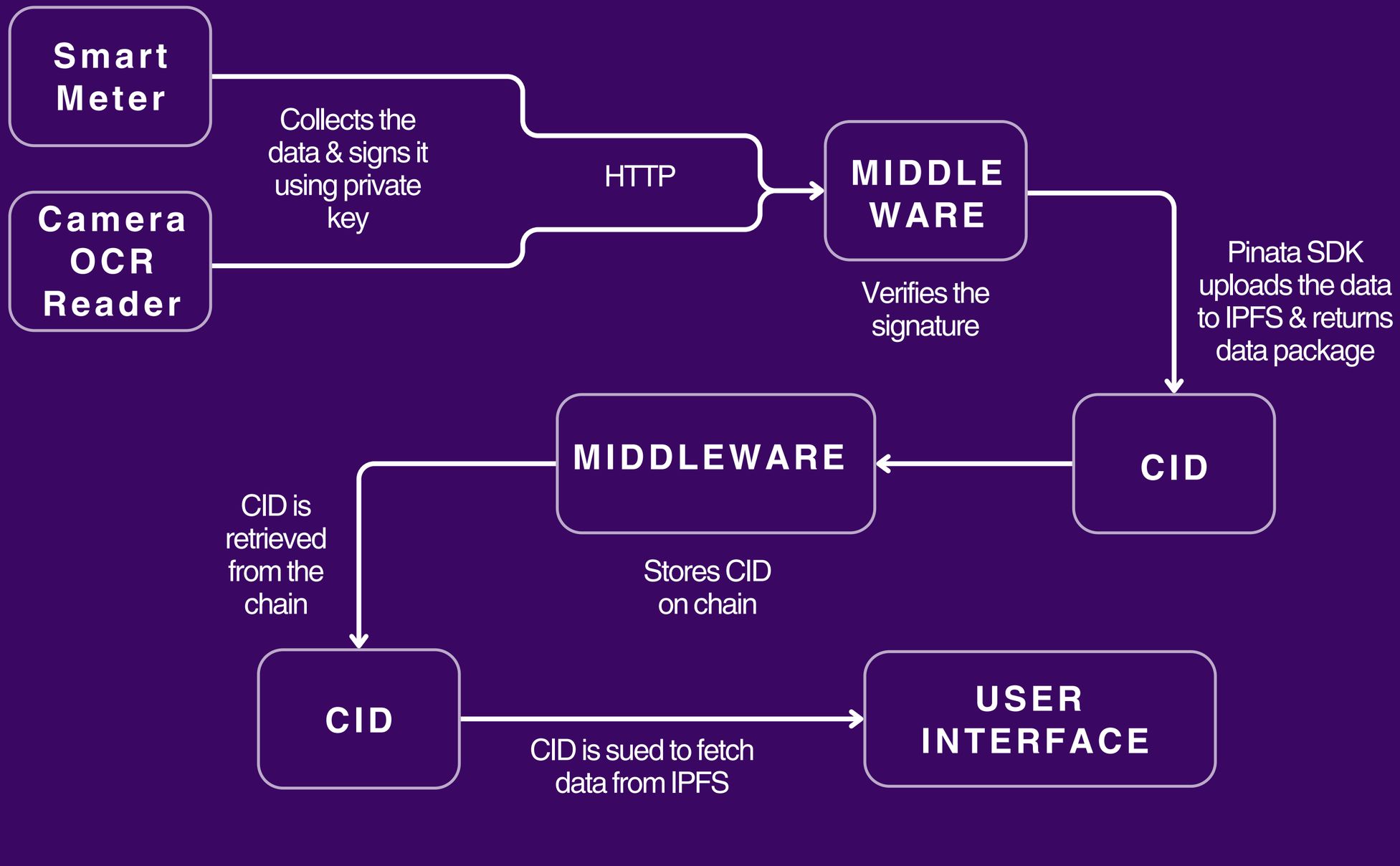
DATA RETRIEVAL PROCESS

To retrieve and access the data:

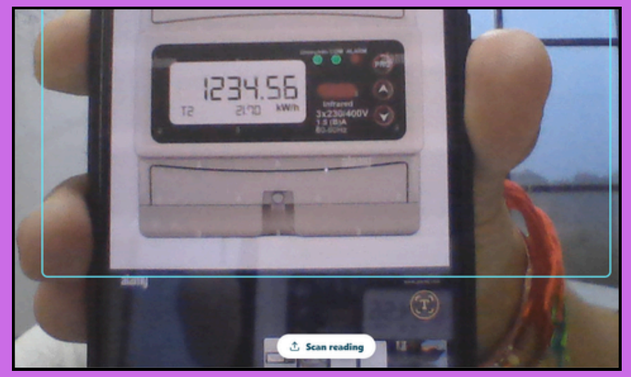
1. Fetch the CID: Retrieve the hash stored on the blockchain.
2. Retrieve Data: Use the CID to fetch the actual JSON data file from **IPFS.

For now we are using mock data for IoT which is getting updated in the real-time but our plan is collecting the data from IoT devices and updating it in on a fixed timeframe like everyday, which resolves the problem of high gas fees.

IOT INTEGRATION WORKFLOW



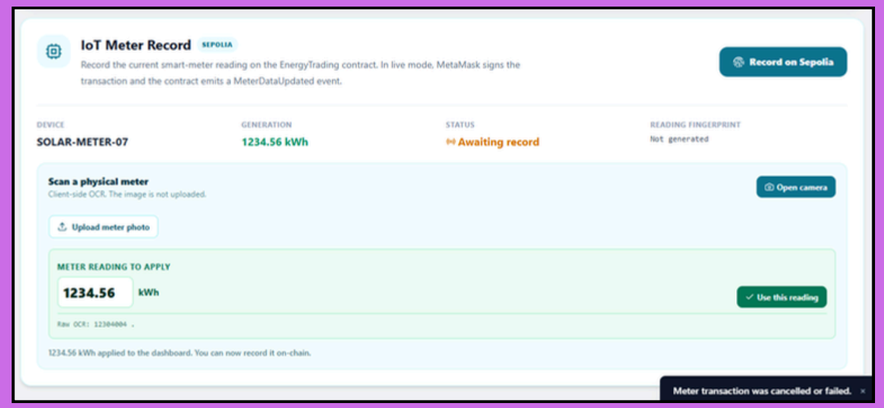
PRODUCT WORKFLOW



Feature 1

Camera Meter Scanner

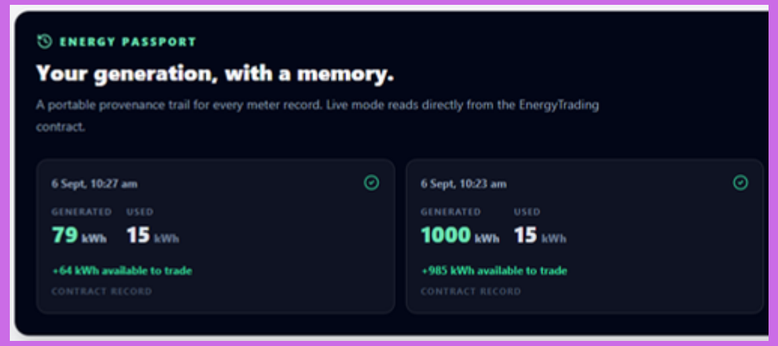
Browser-based local OCR directly extracts kWh values from phone cameras. Users manually confirm the extraction before React state updates.



Feature 2

IoT Meter Records

Once confirmed, live Sepolia smart contracts permanently map historical SmartMeterData to the specific user's connected wallet address.

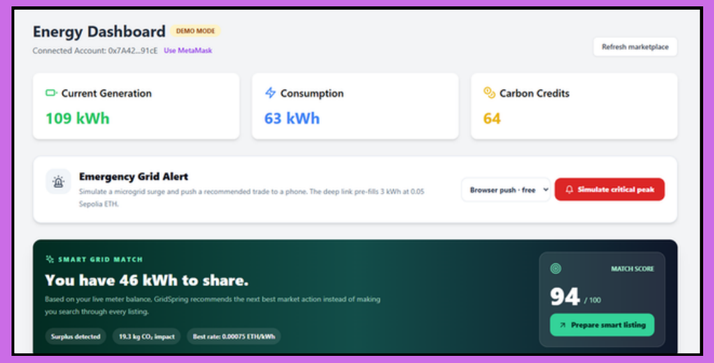


Feature 3

Energy Passport

A visual provenance timeline tracking physical readings, consumption vs generation, and bridging the physical history to digital surplus.

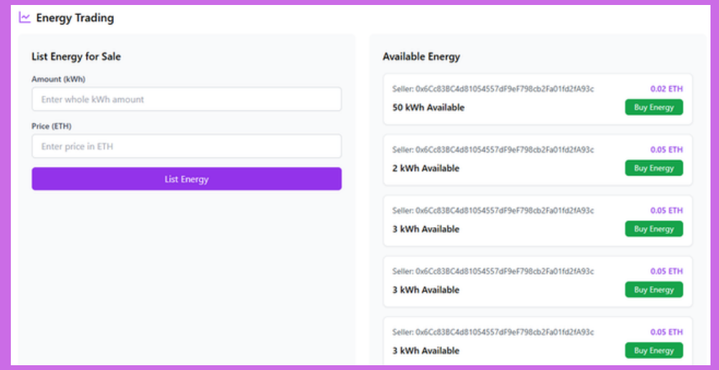
PRODUCT WORKFLOW



Feature 4

Access Modes

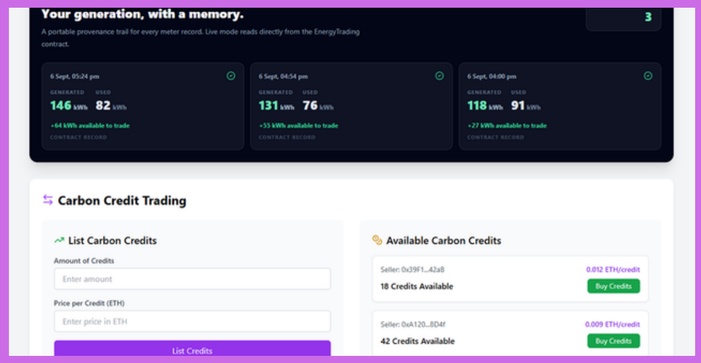
Present instantly via Demo Mode (simulated local state), or connect MetaMask via ethers.js for executing live Sepolia blockchain transactions.



Feature 5

Multi-Listing Market

Energy listings utilize unique numeric IDs, allowing a single wallet to publish and manage multiple independent surplus listings simultaneously.

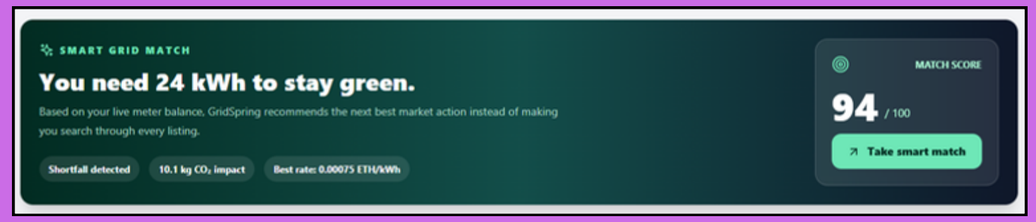


Feature 6

Carbon Credits

Purchasing energy automatically rewards buyers with Carbon Credits (kWh/100). These credits enter a secondary marketplace for independent trading.

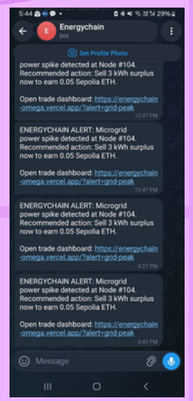
PRODUCT WORKFLOW



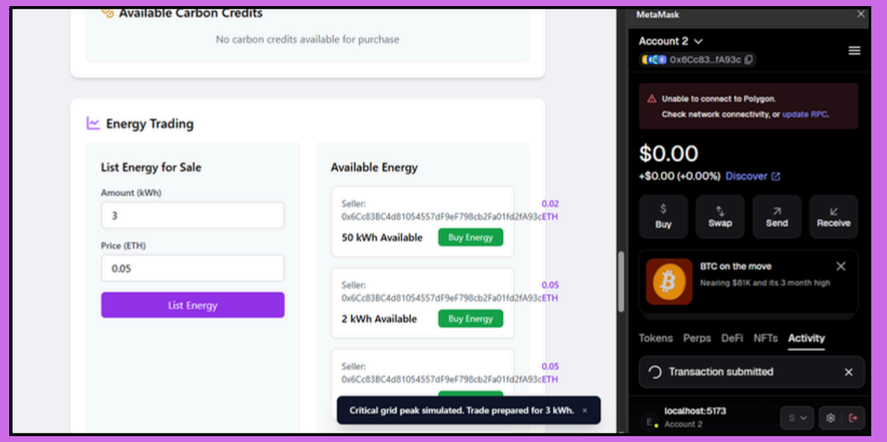
Feature 7

Smart Grid Match

An automated intelligence agent compares real-time generation against consumption. It automatically recommends and pre-fills surplus listings, or automatically searches the open marketplace to find the best available rates to cover power shortfalls.



The dashboard includes a critical-peak simulator. It prepares a valid whole-kWh recommendation of 3 kWh at 0.05 Sepolia ETH and records the emergency event in dashboard activity.
The alert contains a deep link. Opening it activates the dashboard emergency state and pre-fills the trade form





Feature 8

Emergency Grid Alerts

A built-in critical peak simulator triggers emergency conditions, immediately pushing alerts via Browser Notifications or Telegram Bot API. These alerts contain integrated deep links that instantly activate dashboard emergency states and pre-fill critical trade forms.

ZK-SNARKS ARCHITECTURE & RESEARCH

- **Circom & Groth16 Research Setup:** Implemented prototype circuits under zk_snarks/ to test privacy-preserving proofs for smart-meter and energy assets.
- **Positive-Balance Demonstration:** Built a working proof that verifies a user holds a valid, positive balance without exposing their exact energy reserve.
- **Fraud & Double-Spending Prevention:** Uses cryptographic proofs to verify asset ownership off-chain, preventing double-spending of carbon credits or energy tokens while maintaining decentralization.
- **Private Threshold Verification (Roadmap):** Designing production circuits to prove a meter reading exceeds a trading threshold without revealing confidential consumption data.
- **Integrity & Trust Boundary**: Requires trusted signed meter inputs to establish original reading authenticity before generating zero-knowledge proofs.

SYSTEM IMPACT & PRACTICAL USEFULNESS

EnergyChain transforms isolated smart meters into an auditable, peer-to-peer microgrid market that cuts energy costs, automates grid balancing, and provides verifiable carbon offsets.

P2P Energy Monetization & Cost Reduction

- Prosumer Independence: Allows solar producers to sell surplus generation directly to neighbors via Sepolia smart contracts, eliminating centralized utility markups.
- Gas-Optimized Settlements: Uses off-chain IPFS storage and batch hash anchors to prevent prohibitive blockchain transaction fees.

Automated Grid Stability & Smart Matching

- Smart Grid Match Agent: Analyzes real-time consumption vs. generation to recommend optimal buy/sell actions automatically.
- Critical Peak Response: Pushes emergency alerts (via Browser Push or Telegram) with deep links that pre-fill emergency trades during microgrid surges.

Auditable Provenance & Carbon Tracking

- Energy Passport: Maintains an immutable historical trail of meter readings tied to wallet addresses.
- Automated Offsets: Automatically calculates and issues carbon credits on energy trades, creating a transparent record of avoided CO2.

IMPACT AT A GLANCE

86.4Kg
CO2 Avoided

64%
Renewable Share

94/100
Match Score

THANKYOU



RESEARCH



DEPLOYED WEBSITE LINK